

Room-Level CFD Validation of a Hybrid Marine-Air / Seawater-Cooled Pier Data Center

Vishnu Sai Sharan Ankathi • Independent research • September 2026 • Ansys Icepak (AEDT 2026 R1), SolidWorks 2026

270 kW

IT load, 18 racks

21.0 °C

cold-aisle air

33.3 °C

max hot-aisle air

~145

iterations to converge

1. Summary

This report documents the first CFD test of the server room inside a hybrid-cooled, above-water data center concept for the San Francisco Bay. The design uses the region's marine-layer air as the primary cooling source through an indirect air-to-air heat exchanger, with pumped bay water as a secondary stage and a mechanical chiller as a rare fallback. Test 1 isolates the room: the upstream equipment (filters, ducts, both heat exchangers) is represented by its output, 21 °C supply air, so the question answered here is whether that air keeps 270 kW of racks within ASHRAE limits without hot/cold mixing.

Result: yes. The solve converged cleanly, the cold aisle held at 21 °C along almost its full length, hot-aisle air peaked at 33.3 °C (hand calculation: 32.1 °C), and hot and cold streams stayed separated. One real design issue was found: hot air leaks around the row ends through the 195 mm end gaps.

2. Concept and CAD Model

The facility sits on a pile-supported platform above the water. Outside marine-layer air enters through three filter banks; bay water is pumped through a separate loop. Both feed two independent cooling paths, one per rack row, each with an air-to-air exchanger followed by an air-to-seawater exchanger. Room air never mixes with outside air, which matters because Bay air averages ~79% relative humidity.

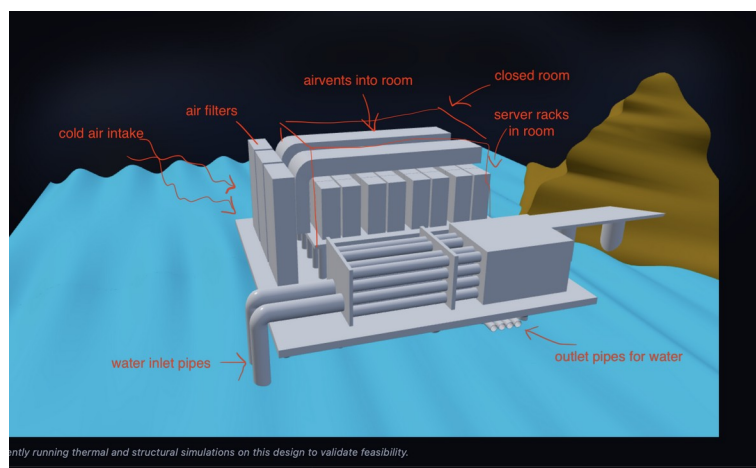


Figure 1. Early concept render with annotated air and water paths.

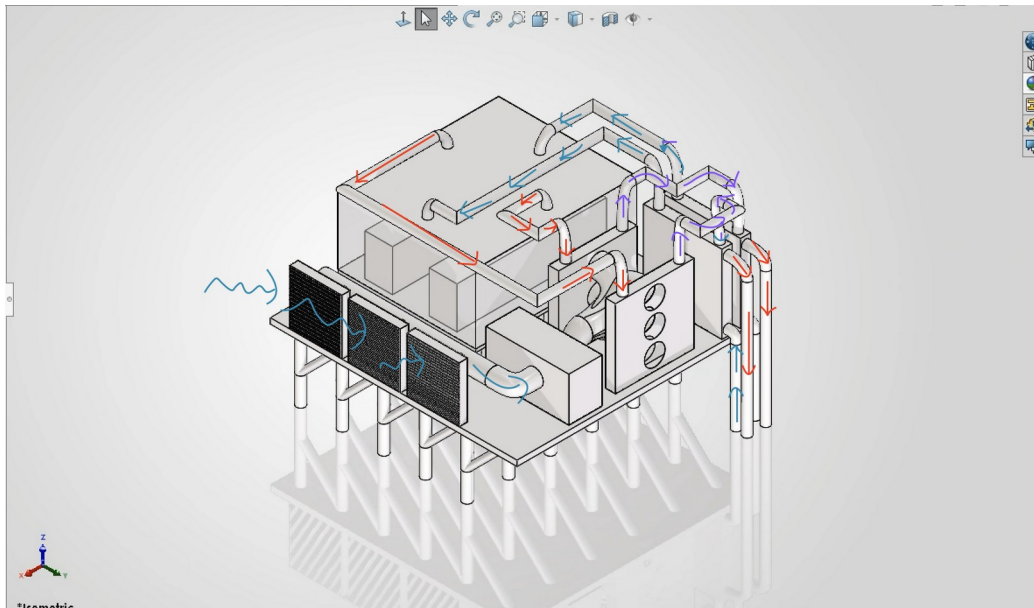


Figure 2. SolidWorks assembly: filter banks, combined manifold, two exchanger paths, sealed server room, seawater piping and pilings.

3. Test 1 Setup

3.1 Scope and geometry

Only the sealed server room was simulated. The full assembly was too large and merged for Icepak (13.7 m × 19.4 m, outside the solver's expected range), so the room was rebuilt natively in AEDT from the design dimensions.

Item	Value
Room (air region)	5790 × 4965 × 2500 mm
Racks	18 (2 rows × 9), each 600 × 1000 × 1500 mm; modeled as one block per row (5400 × 1000 × 1500 mm)
Aisles	Cold aisle 1200 mm (center); hot aisles 882 / 883 mm (sides); 195 mm row-end gaps
Supply opening	Ceiling over cold aisle, 5400 × 1200 mm, velocity inlet -3.1 m/s (Z), 21 °C → 20.1 m ³ /s
Return openings	Ceiling over each hot aisle, pressure (ambient) outlets
Rack model	Recirculating openings per row: 135 kW heat input, 10.06 m ³ /s, extract on cold-aisle face, supply on hot-aisle face

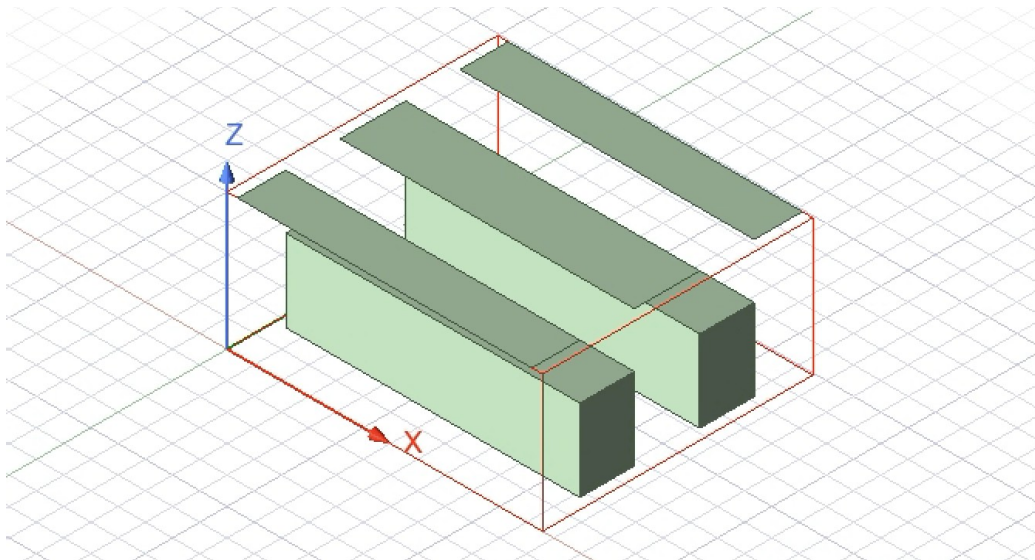


Figure 3. Icepak model: two rack rows, air region, and three ceiling openings (supply center, returns sides).

3.2 Boundary conditions and solver

Setting	Value	Why
Flow regime	Turbulent	~3 m/s through room-scale openings
Gravity	On	Captures buoyant rise of hot-aisle air
Walls	Adiabatic	Isolates rack-to-air heat transfer
Max iterations	300	Solver stops earlier once converged
Mesh	Stair-stepped, max element 100/100/80 mm, min gap 10 mm, ~96,000 elements	Default 1 mm min gap produced failing sliver cells

 The 'Opening Thermal Model' dialog box is shown with the 'General' tab selected. The 'Name' field is 'Supply_ColdAisle'. Under 'Thermal Specification', the 'Temperature' is set to 21 cel and 'External Radiation Temp' is set to AmbientRadTemp. Under 'Flow Specification', the 'Inlet Type' is 'Velocity'. The 'Static Pressure' is set to AmbientPressure. The velocity components are: X Velocity: 0 m_per_sec, Y Velocity: 0 m_per_sec, and Z Velocity: -3.1 m_per_sec. There are 'Use Defaults', 'OK', and 'Cancel' buttons at the bottom.

Figure 4. Supply opening boundary condition: 21 °C, -3.1 m/s vertical velocity.

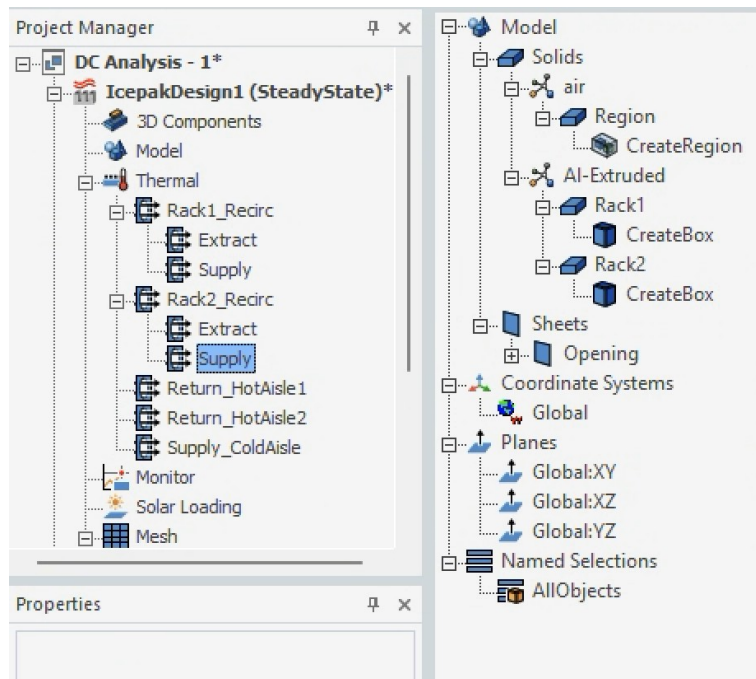


Figure 5. Final boundary list: recirculating rack openings (extract/supply pairs), two returns, one supply.

4. Results

4.1 Convergence and mass balance

The final run converged in about 145 iterations with residuals falling steadily (continuity $\sim 10^{-3}$, velocity $\sim 10^{-5}$, energy $\sim 10^{-7}$). A flow check on the same geometry showed 42,564 CFM entering (design target 42,650 CFM) and an even split between the two returns, so mass balances.

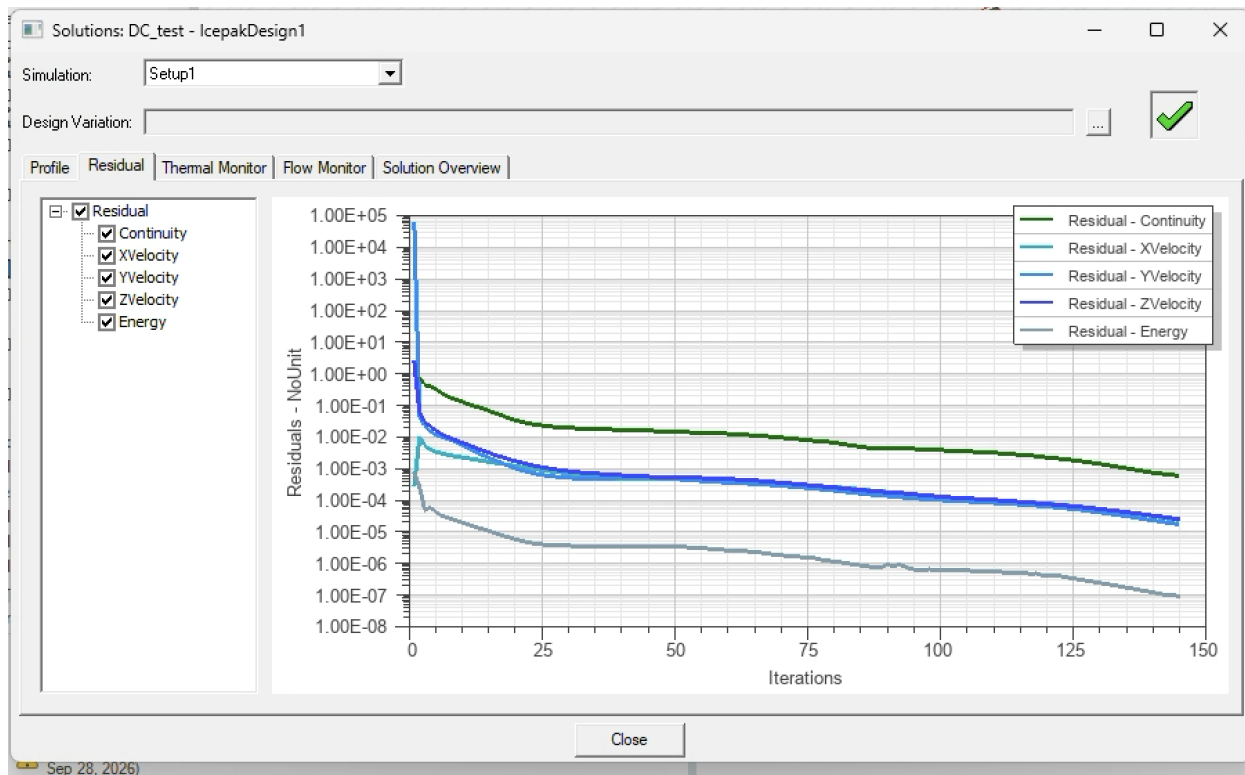


Figure 6. Residual history of the final run, converging well under the 300-iteration cap.

Thermal Flow Fan			
Entity	Mass Flow [kg/s]	Volume Flow [cfm]	Heat Flow [W]
Supply_ColdAisle	23.3302	42564.1	-93787.4
Return_HotAisle1	-11.6624	-21277.2	46883
Return_HotAisle2	-11.6678	-21286.9	46904.5

Figure 7. Flow balance at the openings: supply in, evenly split return out.

4.2 Temperature

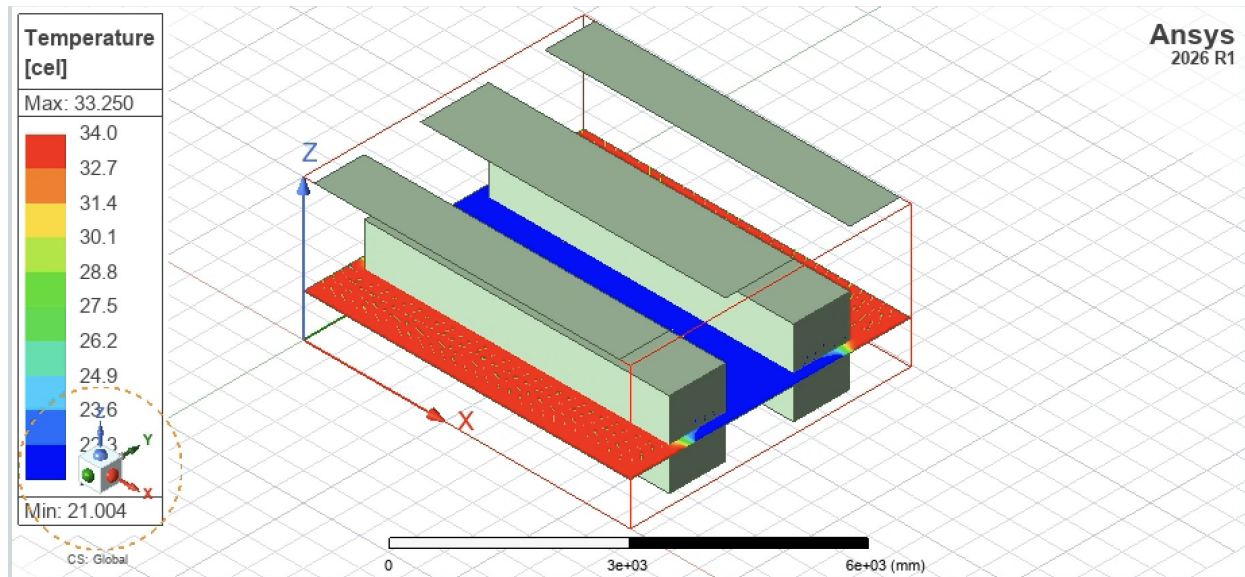


Figure 8. Temperature at rack height: cold aisle at 21 °C, hot aisles ~33 °C, with warm leakage visible at the row ends.

Check	Expected	Simulated
Cold-aisle (rack inlet) air	21 °C	21.0 °C
Hot-aisle air	32.1 °C ($\Delta T = Q / \rho \cdot c_p \cdot V \approx 11.1$ °C)	33.25 °C
ASHRAE recommended inlet	18–27 °C	Met (21 °C)

4.3 Airflow

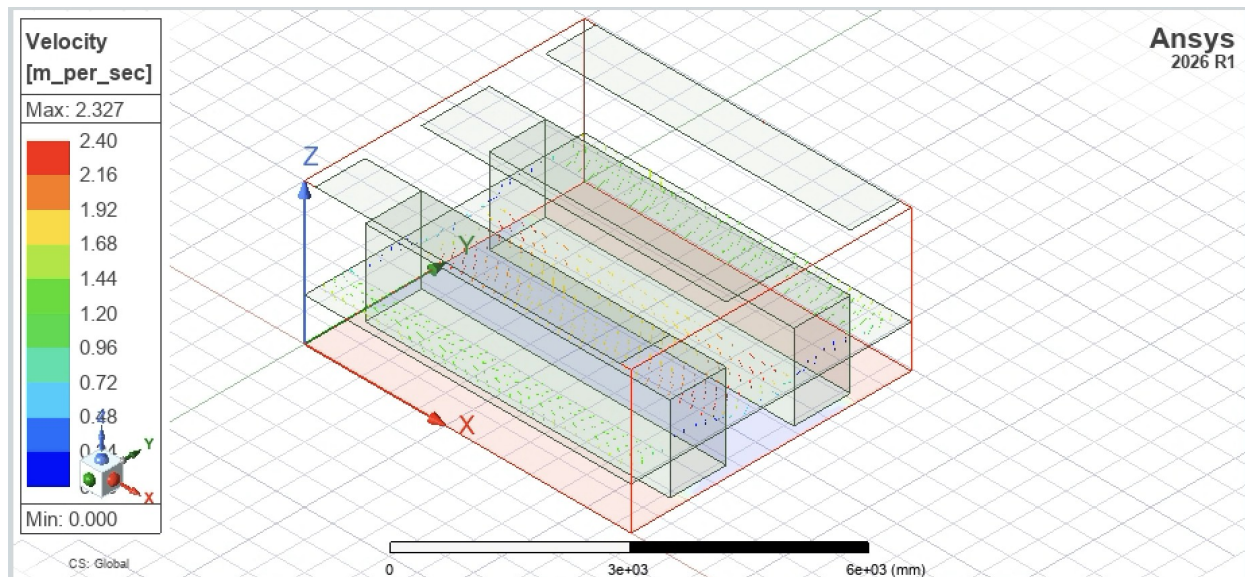


Figure 9. Velocity vectors on a mid-height slice: max 2.33 m/s, air falling into the cold aisle and moving out through the racks.

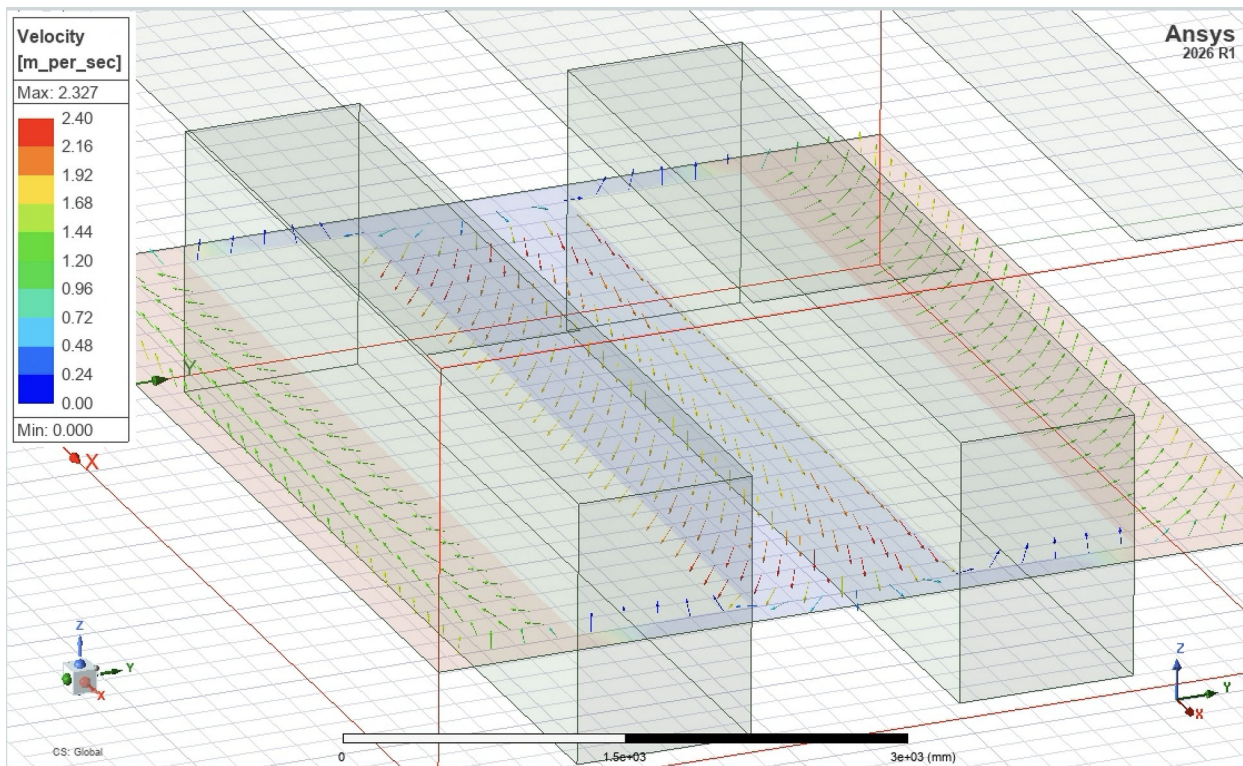


Figure 10. Close-up of the airflow: downward flow in the cold aisle, turning outward through both rack rows toward the hot aisles.

5. Findings

- **Aisle separation works.** The hot/cold containment layout keeps rack inlets at supply temperature across nearly the full row length.
- **CFD agrees with the hand calculation** to about 1 °C on hot-aisle temperature.
- **Row-end recirculation is the main weakness.** Hot air wraps around the 195 mm row-end gaps into the cold aisle. End-of-row panels or doors are the standard fix and are the next test.
- **Honest caveat:** the recirculating rack model imposes the rack airflow and heat, so the ~12 °C rise across each rack is an input. What the CFD adds is how the room air behaves around the racks.

6. Troubleshooting Log

Getting to a valid run took several iterations; the lessons are worth keeping for future models.

Problem	Cause	Fix
Model dimension warning	Full platform assembly imported as one merged solid	Rebuilt room natively in AEDT
Region padding rejected	Absolute padding means world coordinates, not offsets	Set -X/-Y/-Z = 0 first, then + faces to room size
Opening creation failed	Opening needs a selected face / sheet	Drew boxes, assigned Free Opening
Air temp 1173 °C, velocity 0	Racks as solid heat blocks lose heat only through their skin; plots were on the floor (wall, $v = 0$)	Recirculating rack openings; plotted on mid-air slices

Problem	Cause	Fix
"Poor mesh quality" failure	1 mm minimum gap created sliver cells	Stair-stepped mesh, 10 mm min gap, 100 mm max element
Crashes / lost saves	Project on network drive (VLab)	Save archives (.aedtz) and copies off the network drive

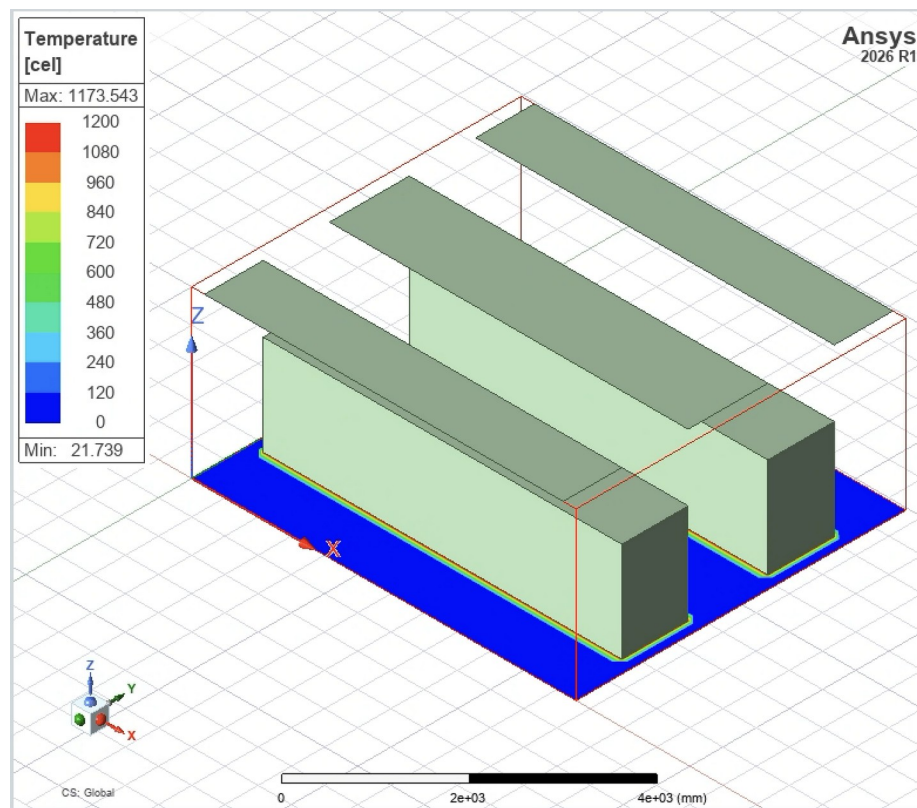


Figure 11. Early failed run: solid-block racks produced a physically impossible 1173 °C before the rack model was corrected.

7. Next Steps

- Test 2: add end-of-row panels and rerun; compare row-end temperatures.
- Test 3: summer stress case with 25 °C supply air; confirm rack inlets stay ≤ 27 °C.
- Heat exchanger check (ϵ -NTU) for both stages and fan-power estimate from filter, duct and coil pressure drops.
- Year-round control simulation (Simulink or Python) of the air $\rightarrow$ seawater $\rightarrow$ chiller sequencing using hourly Bay climate data.

Modeling, setup and design decisions by the author. AI tools were used to assist with calculations, troubleshooting and report drafting.